FINAL REPORT

Submission date: 12/08/2023

ABSTRACT

This report has been prepared to document activities from October through December in the Product Life Cycle Management course. This report also contains a detailed view of the RYOBI 18V handheld sprayer, drawings of original and modified part, change management, 3D printed parts, simulation and modeling and data analysis.

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Tarun Bayya

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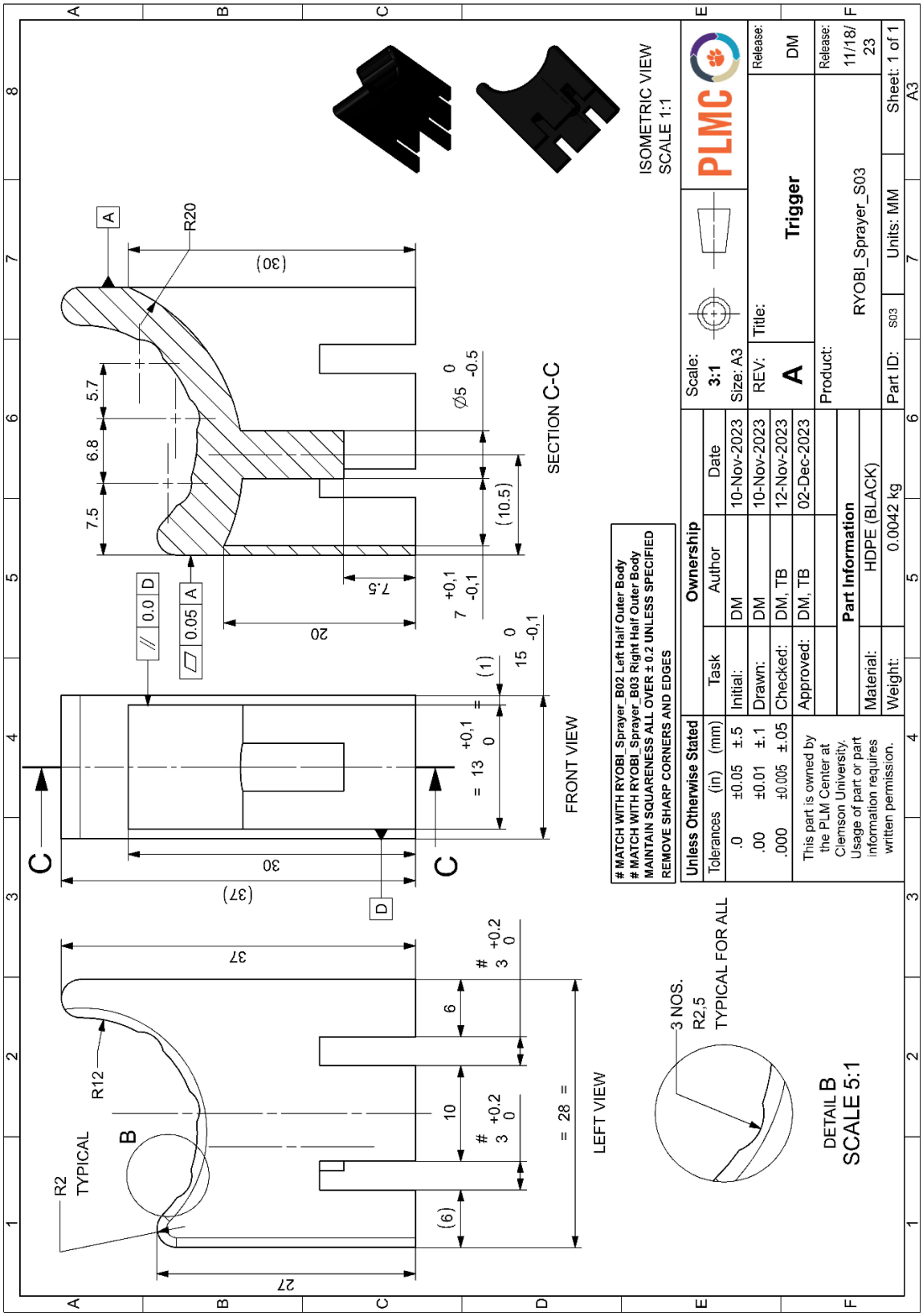
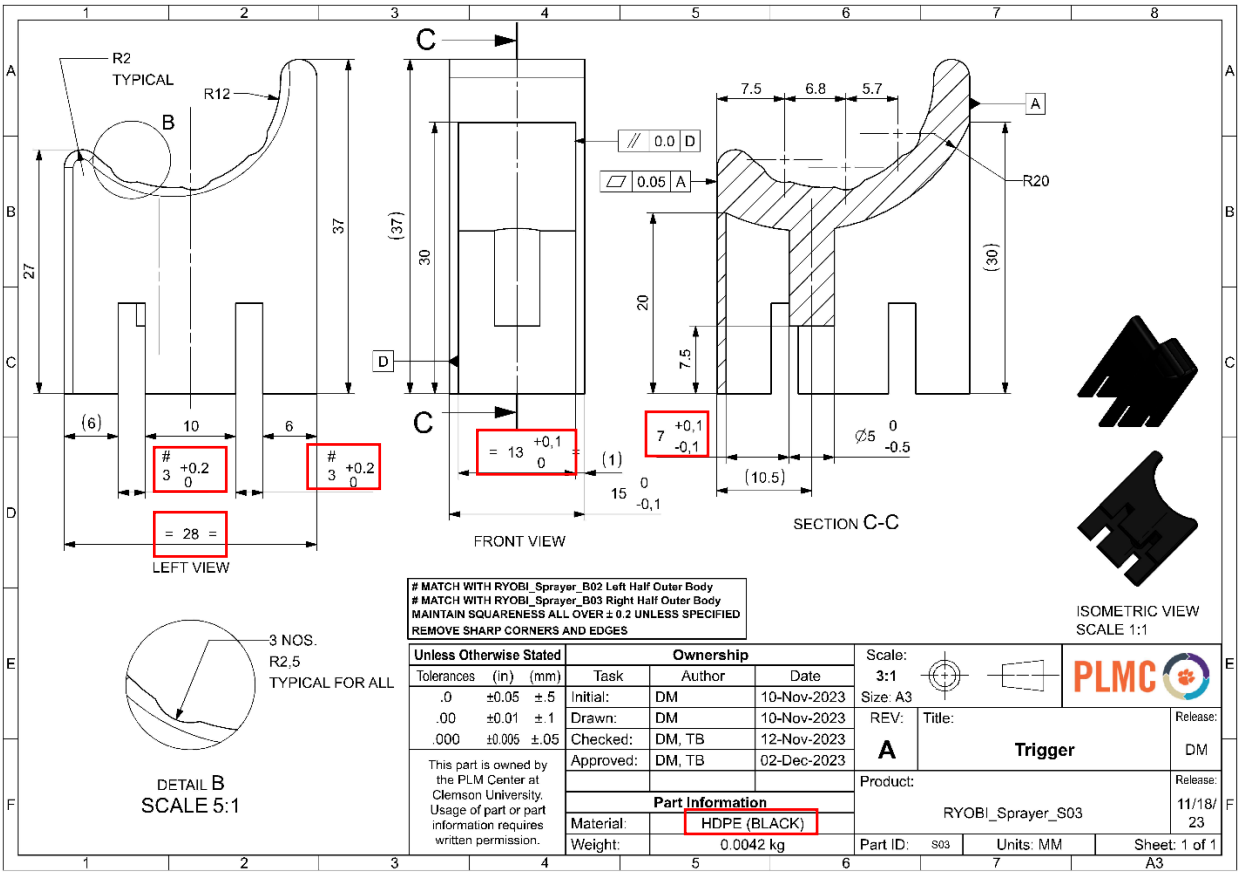
A. Technical Documentation



Tolerances' time, costs and reasons



Essential Details	Values	Feature Nos/Datum	Tolerances	Verification Time (s)	Time for Verification/ year (s)	Cost for Verification ($)
Shaft Location	7	1	±0.01	3	3600	26
Slot (Gap)	3	4	+0.02	9	10,800	78
Color	Black	1	Visible Black	1	1200	9
Parallel Edges	13	D	±0	12	14,400	104
Flat Sides	28	A	+0.05	8	9600	70
Total				33	39,600	287

When designing the trigger for our product we took care in selecting tolerances based on a thorough evaluation of functional requirements and the surrounding geometry. Each specification was meticulously chosen to ensure optimal performance and reliability while maintaining a balance between precision and economic viability.

For the shaft location, the tolerance of ±0.01 was chosen to guarantee accurate alignment and optimal functioning of the trigger within the larger system. As, if the location of the shaft was misaligned with the switch, the trigger would not be able to push onto the switch appropriately. The verification time of 3 seconds and associated annual cost of $26 strike a reasonable compromise between thorough quality checks and efficient production processes. The slot (gap) tolerance of +0.02 was determined to allow for ease of assembly and prevent interference issues. The slot ensures that the trigger stays level and has enough space to glide smoothly without interference. The verification time of 9 seconds

and associated annual cost of $78 reflects the need for a more detailed inspection due to the increased complexity of the four slots.

The decision to opt for a black color specification is based on maintaining color scheme for the product line. The associated verification time and cost are minimal at 1 second and $9, respectively, given the simplicity of visual inspection. The parallel edges, with a tight tolerance of ±0, are crucial for maintaining the operational portion of the trigger. If the sides of the trigger are not parallel, then the trigger would not sit perfectly into the sprayer and can potentially have jamming issues. The verification time of 12 seconds and annual cost of $104 are justified by the critical nature of this feature for the overall functionality. Lastly, the flat sides with a tolerance of +0.05 were chosen to balance manufacturing feasibility and cost-effectiveness, and overall look. The associated verification time of 8 seconds and annual cost of $70 were determined to be acceptable for ensuring functionality without overburdening the production process.

B. Engineering Change Management

Completed ECO

Title of ECO	Trigger Deployment Ergonomics and Safety Challenges	ECO ID	ECO - 00693
Requesting Person	John Morris	Submitted Date	20 NOV 2023
Product Manager	Dr. John Wagner	Date needed	29 NOV 2023
Owner	Dhwanil Mehta, Tarun Bayya	Urgency Level	Normal
Affected Products	RYOBI PSP01B ONE+ 18V Cordless Handheld Sprayer		
Issue	• Finger fatigue while deploying the trigger. • Finger slipping off the trigger while trigger is deployed. • Loss of control when trigger is wet.		
Description	There are 2 product updates affecting 1 part and creating 1 additional part: 1. Trigger needs to be longer. 2. New die for trigger needs to be similarly made. 3. New Part: Existing trigger cannot be modified, entire new part production required.		
Part(s) Impacted	Trigger		
Associated Part Files	RYOBI_Sprayer_S03 Trigger.prt		
Status of ECO	Completed		
Requesting Person	Dhwanil Mehta, Tarun Bayya		
Date of ECO Completion	28 NOV 2023		
Date of Modification(s)	22 NOV 2023; 24 NOV 2023; 28 NOV 2023		

Concepts Generated

Concepts Generated for solving the issues mentioned in ECO.
1. Implement a textured surface on the trigger for improved grip.
2. Integrate a lock on the trigger part for enhanced safety features and reduce fatigue.
3. Increase size of the trigger by its width for more contact surface area.
4. Construct the trigger from a softer material to enhance user comfort and cushioning of finger better.
5. Utilize Geckskin material on the trigger for enhanced tactile feedback.
6. Incorporate a finger indent on the trigger for better ergonomics.
7. Increase the length of the trigger to accommodate the use of multiple fingers.
8. Knurling texture on the trigger, diamond shaped and indented groove.

Selection process for the optimal concept

The scores for the concept are determined based on how the suggested modification for triggering works taking into account practical factors, like cost, time and disturbance during implementation. Performance assesses the impact on trigger functionality, while Appeal gauges the subjective attractiveness of the design. The scores for each criterion are multiplied by their respective weights, and the sum yields the overall score. The cost has been assumed, based on internet searches and discussions within the group on the score. The time and disturbance have been scored using fair judgement and discussions. As for Performance, the group have decided the scores based on created imaginary scenarios, placing we from the customer's perspective trying to gauge the performance of the concepts. Appeal has been rated based on individual member's vote and averaging the value achieved by these discussions.

Concept	Cost (0.2)	Time (0.1)	Disturbance (0.2)	Performance (0.3)	Appeal (0.2)	Score
Implement a textured surface on the trigger for improved grip.	2	1	1	2	4	2.1
Utilize Geckskin material on the trigger for enhanced tactile feedback.	4	1	1	2	3	2.3
Incorporate a finger indent on the trigger for better ergonomics.	2	2	3	4	2	2.8
Increase the length of the trigger to accommodate the use of multiple fingers.	3	3	3	5	1	3.2
Increase size of the trigger by its width for more contact surface area.	2	3	3	3	4	3

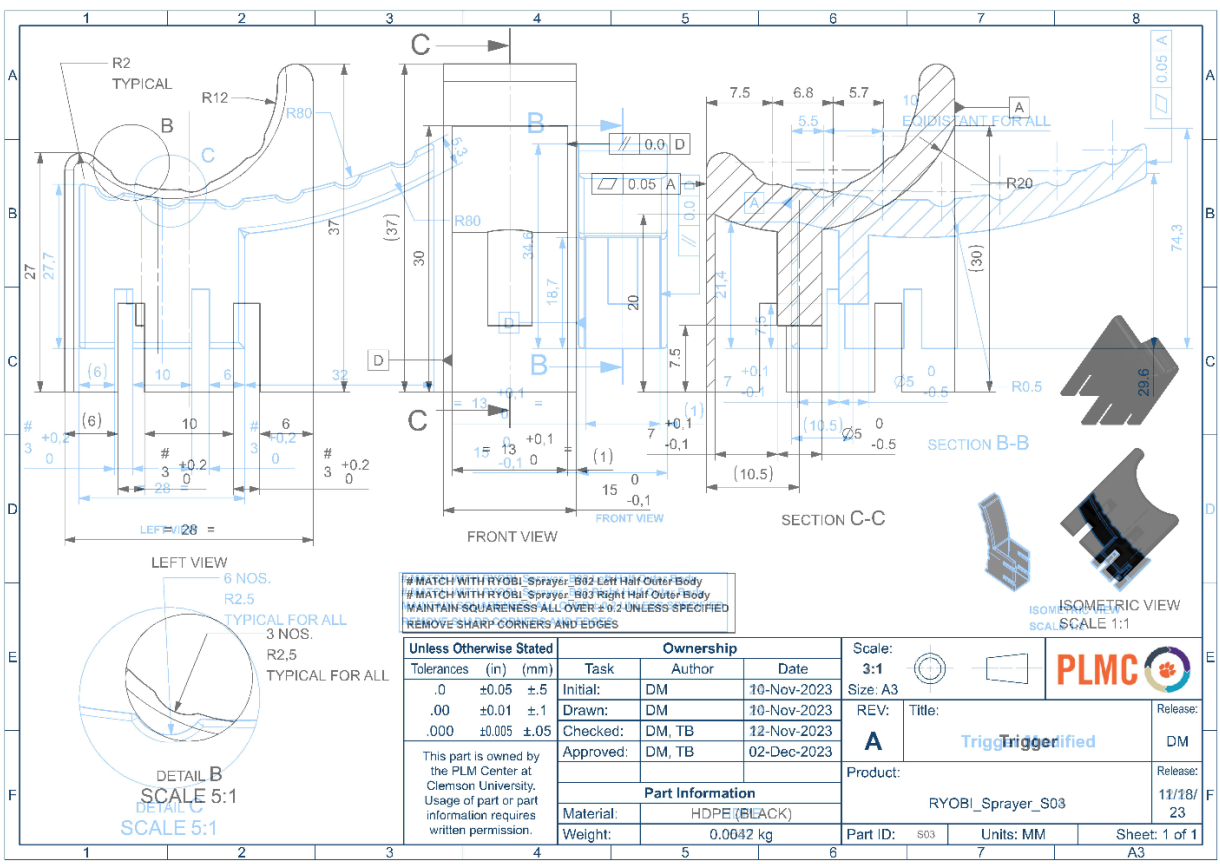
Explanation of Chosen concept

The concept that we have selected is to make the trigger longer so that it can be used with fingers providing an experience and more comfort. This modification would enhance the trigger's functionality by enabling users to utilize more than one fingers when pushing onto the trigger. The longer trigger spreads out the force, across the fingers making it easier to aim and keeping your hand in a position. It also offers a space, for engagement making it ideal for individuals with hand sizes and grips. This modification would thus help reduce fatigue and prevents slipping in wet weather conditions, and also help reduce finger fatigue and slippage. It is also noteworthy to mention that the redesigned trigger comfortably fits three fingers.

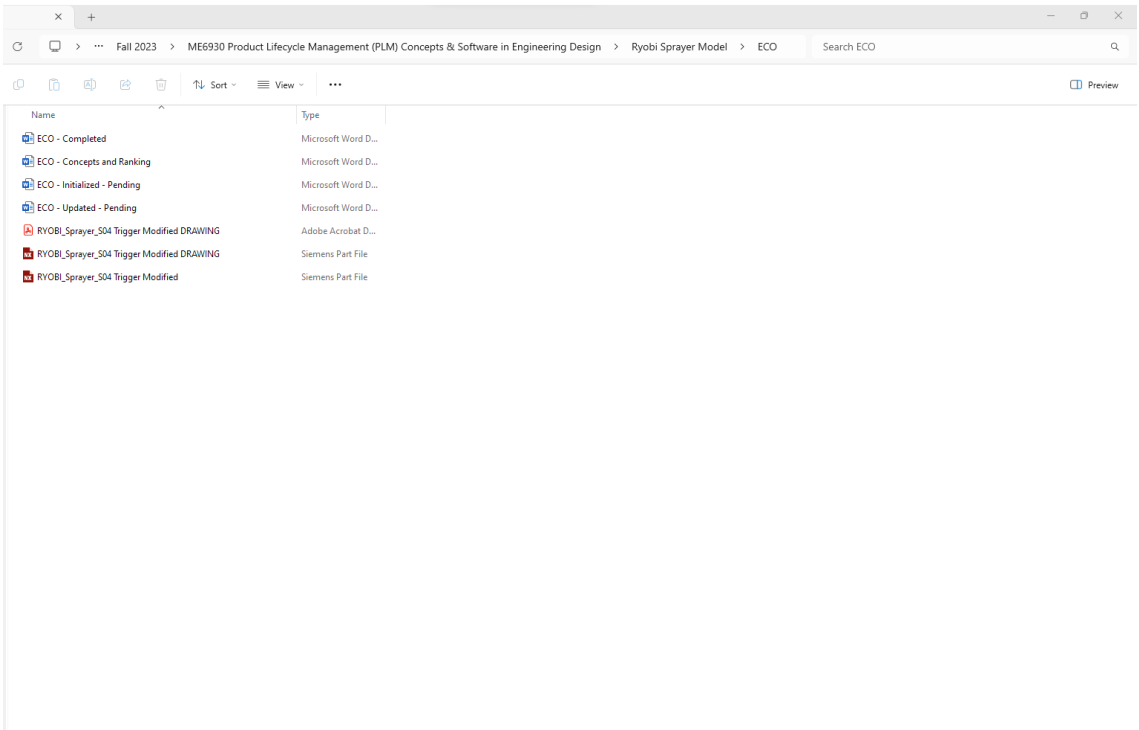
Actual and estimated work for modifications

Title: Trigger Deployment Ergonomics and Safety Challenges							
TASK	Description	Owner	Date of Completion	Estimated Time (s)	Estimated Cost (35$/hour) ($)	Actual Time (s)	Estimated Cost (35$/hour) ($)
	Establish optimal length for trigger	Dhwanil Mehta, Tarun Bayya	22 NOV 2023	600	6	300	3
	Determine optimal design for new trigger	Dhwanil Mehta, Tarun Bayya	22 NOV 2023	600	6	300	3
	Document initial design on paper	Dhwanil Mehta, Tarun Bayya	24 NOV 2023	900	9	120	2
	Modify design in NX	Dhwanil Mehta	25 NOV 2023	3600	35	3600	35
	Develop engineering drawings of modified trigger	Dhwanil Mehta	28 NOV 2023	2700	27	3600	35
	Setup STL file for 3D printing	Dhwanil Mehta	28 NOV 2023	900	9	600	6

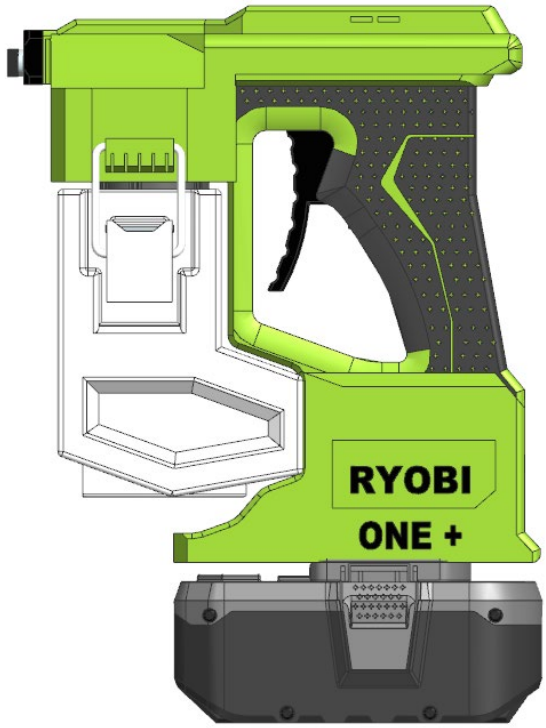
Original vs Modified Technical Drawing



ECO data storing system

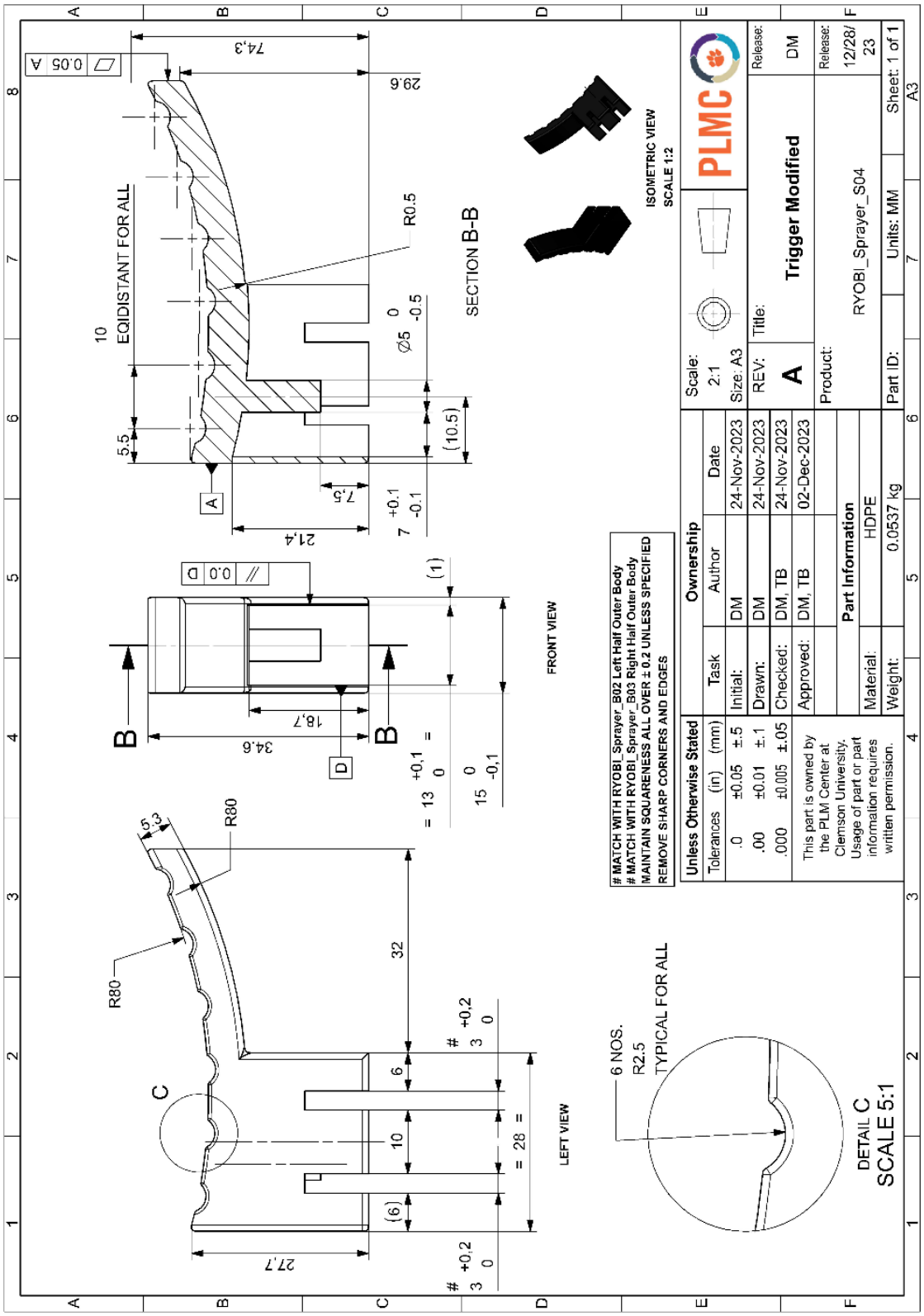


Updated assembly of the sprayer with modified trigger



C. Additive Manufacturing

Technical drawing for modified trigger



Tolerances' time and costs

Essential Details	Values	Feature Nos/Datum	Tolerances	Verification Time (s)	Time for Verification/ year (s)	Cost for Verification ($)
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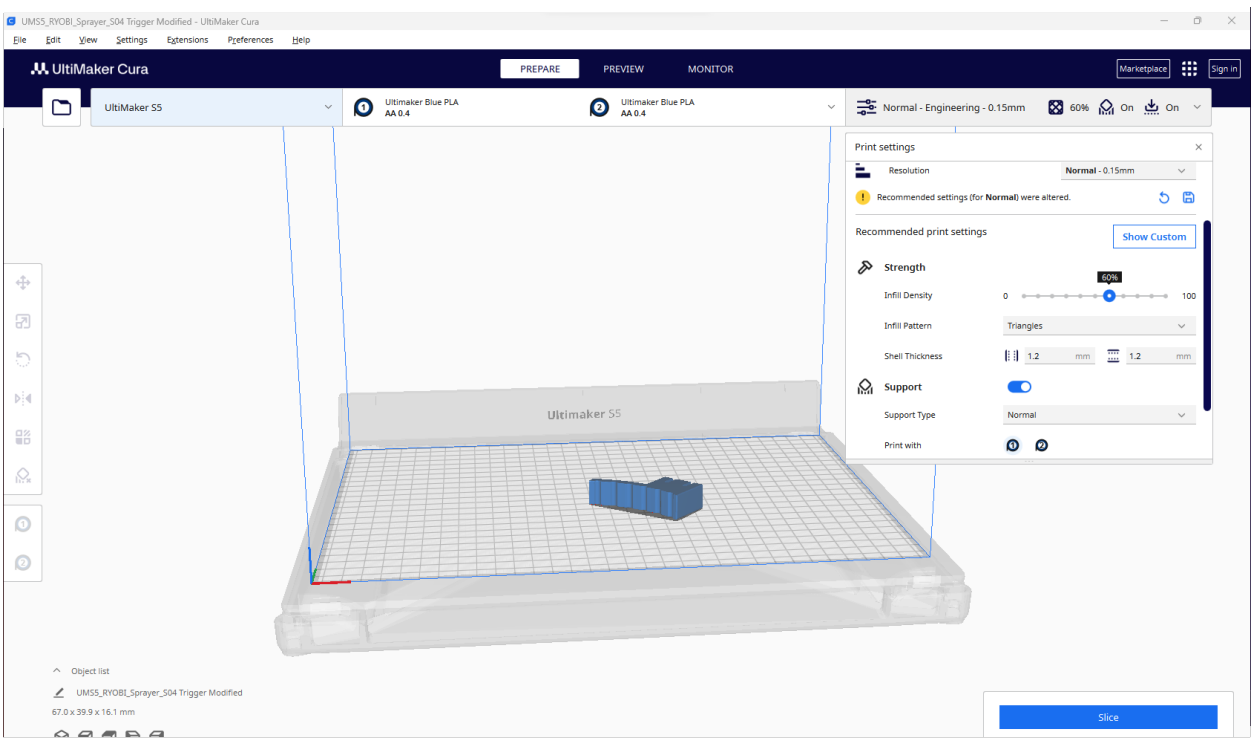
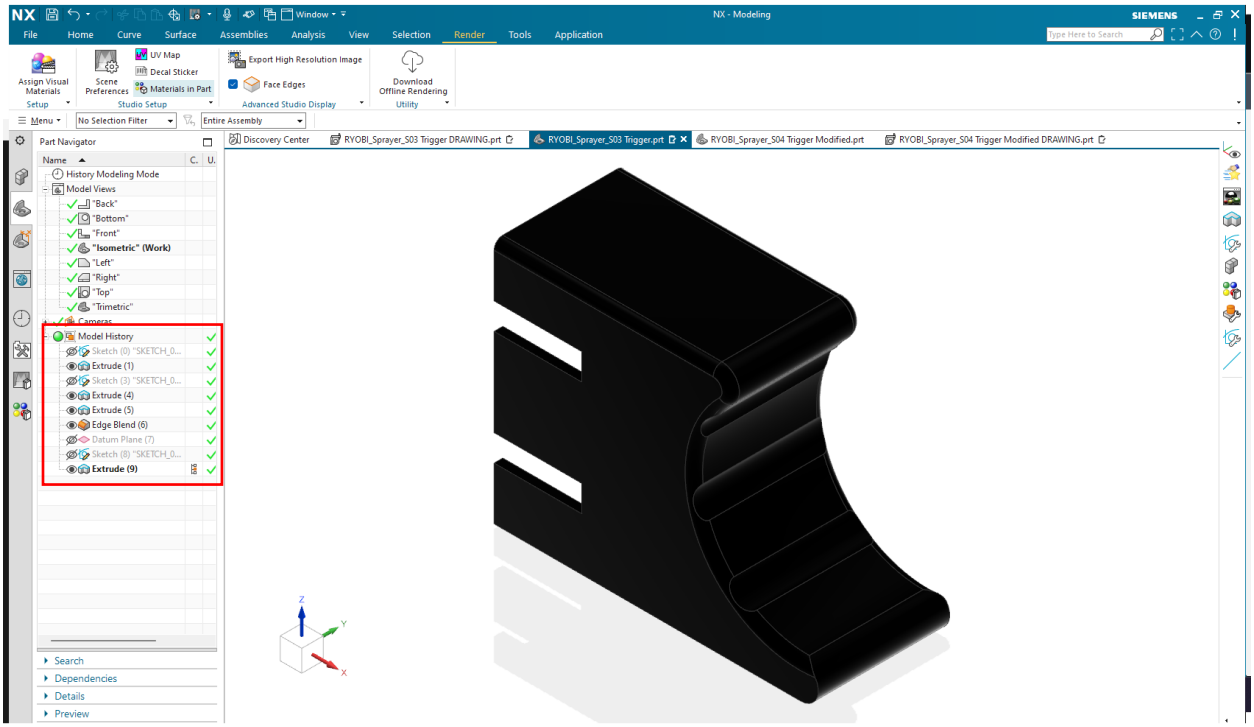
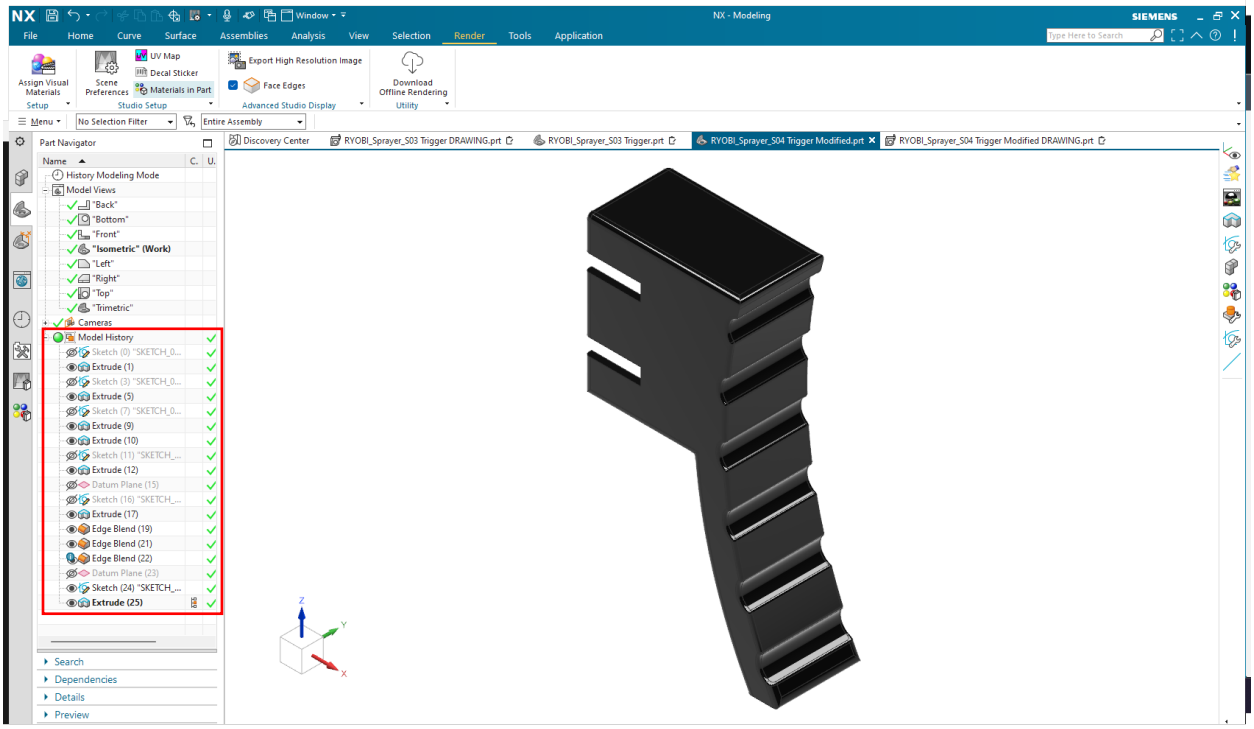
Color	Black	1	Visible Black	1	1200	9
Parallel Edges	13	D	±0	12	14,400	104
Flat Sides	28	A	0.05	8	9600	70
Height	32	1	-	4	4800	35
Total				37	44,400	322

Technical settings in UltiMaker Cura



Comparison and modelling changes to original and new trigger





Changes to new model		
Features Unchanged	**Deleted**	**New/Added Changes**
Sketch (0) "SKETCH_000"	Extrude (4)	Sketch (7) "SKETCH_003"
Extrude (1)	Edge Blend (6)	Extrude (10)
Sketch (3) "SKETCH_001"	Datum Plane (7)	Sketch (11) "SKETCH_005"
Extrude (5)	Sketch (8) "SKETCH_003"	Extrude (12)
Extrude (9)		Datum Plane (15)
		Sketch (16) "SKETCH_006"
		Extrude (17)
		Edge Blend (19)
		Edge Blend (21)
		Edge Blend (22)
		Datum Plane (23)
		Sketch (24) "SKETCH_007"
		Extrude (25)

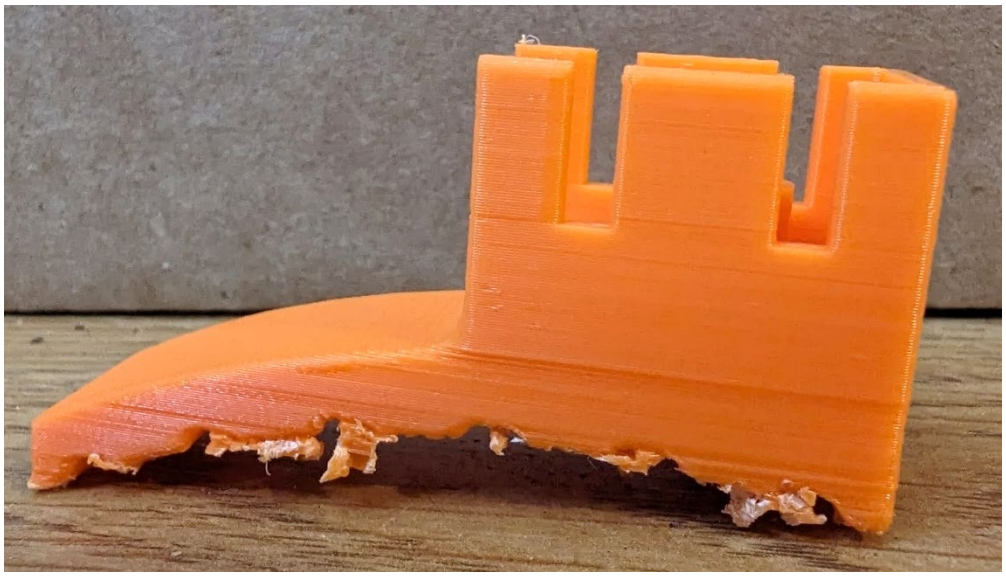
3D printed prototype, validation and next steps

The model of the prototype printed was first assembled and checked using NX modeling software by using assembly features. The model of the modified trigger was assembled into the Sprayer Assembly to validate the given geometry and location. Since we did not have first-hand access to the physical sprayer, we used the imaging capabilities of NX modeling software to perform the validation process. The image of the original trigger was placed next to a scale and oriented in all directions according to the vernier caliper measurements, which were taken physically during the initial

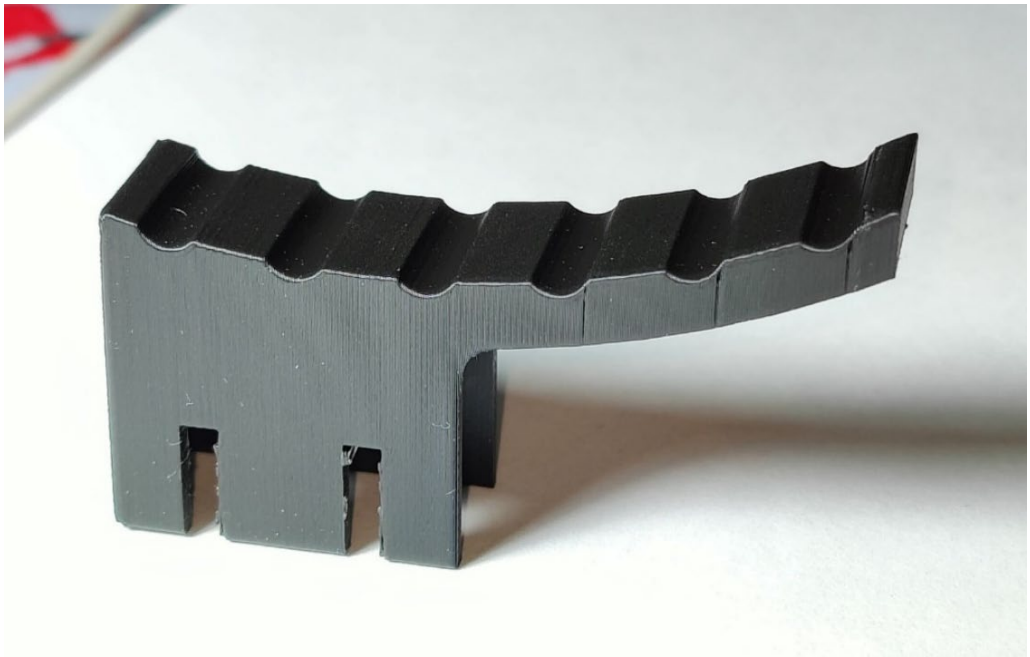
report of the project. Once the scale of the original trigger was set in the software, another picture-printed sprayer was placed into the software and resized according to the dimensions taken physically. These images were placed on top of each other and, using opacity settings in the software, were verified for all views: top, front, back and bottom.

Since the dimensions of the original model were based on the actual dimensions of the physical part, the assembly-specific dimensions matched perfectly in the new model. Through our comprehensive validation process, we could also verify all of the new model's dimensions. However, we could not check the printed geometry by assembling the entire sprayer with the 3D-printed part.

Iteration 1: Improper settings in UltiMaker Cura



Iteration 2: Updated Settings in UltiMaker Cura



Meta Data	
Names of group members	Dhwanil Mehta; Tarun Bayya
Responsibilities of various group members	Design 3D printing setting; 3D print; Verify model
Date	11/30/23
ID of part numbers affected	RYOBI_Sprayer_S03.prt
Version of the report	1.0

D. Modeling and Simulation

Programming Platform and Language for Dynamic Simulation and Agent Based Simulation

We performed the simulation for both cases by using the Python programming language and processing the code using Jupyter Notebook platform. Python is a highly capable programming language that can be used to code and simulate various problems. It is fast, easy and highly adaptable. Conversely Jupyter Notebook offers a user-friendly interface that enables the incorporation of code and comprehensive documentation, within a single document.

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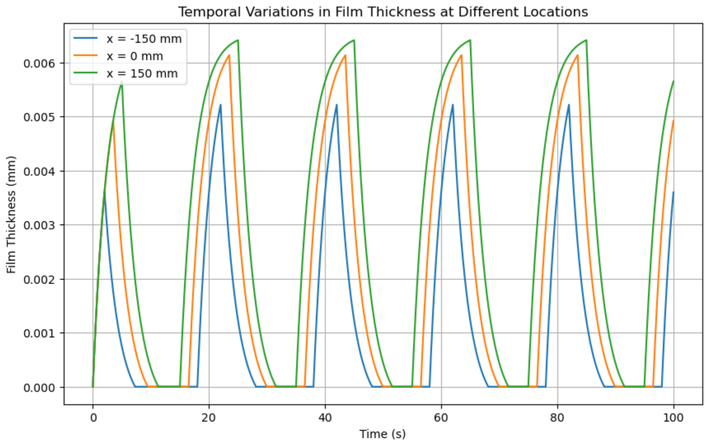
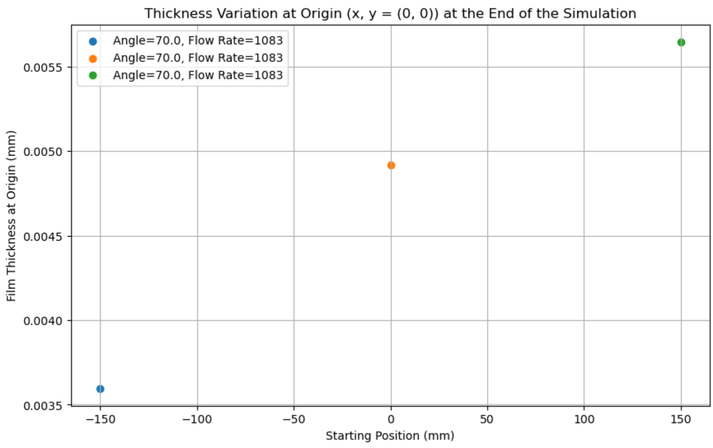
In [2]:

```python
import numpy as np
import matplotlib.pyplot as plt

def calculate_Xt(t, period=10, velocity=100):
    """
    Determine the horizontal displacement at time t for oscillating motion.
    """
    #Assuming the sprayer initiates rightward motion at t=0.
    if int(t / period) % 2 == 0:
        return (t % period) * velocity
    else:
        return period * velocity - (t % period) * velocity

def liquid_height_change(Q, a, T, lambda_, alpha, k, S):
    """
    Determine the variation in liquid depth at position (x, y) over time t.
    """
    A = min(alpha, max(0, alpha * T / k)) #Absorption Rate
    if S: # If undergoing aerosol application.
        return (Q / a) - lambda_ * (T > 0) - A
    else:
        return -lambda_ * (T > 0) - A

def simulate_sprayer():
    # Parameters
    dt = 0.01 # time_interval
    fixed_x_values = np.array([-150, 0, 150]) # position_values
    h = 500 # height in mm
    theta = 7 * np.pi / 18 # spray_angle_rad
    lambda_ = 2e-4 # evaporation rate in mm/s
    alpha = 0.04 # max absorption rate in mm/s
    k = 0.1 # height for max absorption in mm
    a = np.pi * (h * np.tan(theta / 2))**2 # area covered by sprayer

    # Time parameters
    t_final = 100 # simulation_duration
    t_steps = np.arange(0, t_final + dt, dt)  # time_steps

# Initialize surface for all fixed x values
    T = np.zeros((len(fixed_x_values), len(t_steps))) # Evolution of liquid height over successive time intervals at a constant x-coordinate.

    # Iterate through each constant x position.
    for j, x in enumerate(fixed_x_values):
        # Iterate through temporal increments.
        for i in range(len(t_steps) - 1):
            t = t_steps[i]
            X_t = calculate_Xt(t) # Retrieve the present horizontal coordinate (X-axis) of the sprayer.
            Y_t = 0  # Assuming a fixed Y position remains unchanged.
            Q_t = 1083  # consistent volumetric rate in mm^3/s
            S = (x - X_t)**2 <= (a / np.pi)

            # Runge-Kutta 4th order
            k1 = liquid_height_change(Q_t, a, T[j, i], lambda_, alpha, k, S)
            k2 = liquid_height_change(Q_t, a, T[j, i] + 0.5 * k1 * dt, lambda_, alpha, k, S)
            k3 = liquid_height_change(Q_t, a, T[j, i] + 0.5 * k2 * dt, lambda_, alpha, k, S)
            k4 = liquid_height_change(Q_t, a, T[j, i] + k3 * dt, lambda_, alpha, k, S)
            T[j, i+1] = T[j, i] + (dt / 6) * (k1 + 2*k2 + 2*k3 + k4)

    return T
# Plot results for each fixed x location
def plot_results(t_steps, T, fixed_x_values):
    plt.figure(figsize=(10, 6))
    for idx, x_val in enumerate(fixed_x_values):
        plt.plot(t_steps, T[idx, :], label=f'x = {x_val} mm')
    plt.title('Temporal Variations in Film Thickness at Different Locations')
    plt.xlabel('Time (s)')
    plt.ylabel('Film Thickness (mm)')
    plt.legend()
    plt.grid(True)
    plt.show()

# Define a function to plot the thickness variation at the end of the simulation
def plot_thickness_variation(T_values, fixed_x_values, spray_angles, flow_rates):
    plt.figure(figsize=(10, 6))
    for i, x in enumerate(fixed_x_values):
        for j, angle in enumerate(spray_angles):
            for k, flow_rate in enumerate(flow_rates):
                plt.scatter(x, T_values[i, -1], label=f'Angle={np.degrees(angle):.1f}, Flow Rate={flow_rate}')

    plt.title('Thickness Variation at Origin (x, y = (0, 0)) at the End of the Simulation')
    plt.xlabel('Starting Position (mm)')
    plt.ylabel('Film Thickness at Origin (mm)')
    plt.legend()
    plt.grid(True)
    plt.show()

# Define a function to print the thickness table
def print_thickness_table(T_values, fixed_x_values, spray_angles, flow_rates):
    print("Thickness at Origin (x, y = (0, 0)) at the end of the simulation:")
    print("\n{:>20}{}".format("Spray Angle (deg)", "".join(f"{angle:.1f}" for angle in np.degrees(spray_angles))))
    print("-" * 50)

    for i, x in enumerate(fixed_x_values):
        print("{:<20}".format(x), end="")
        for j, angle in enumerate(spray_angles):
            for k, flow_rate in enumerate(flow_rates):
                print("{:>20.2f}".format(T_values[i, -1]), end="")
        print()

# Run the simulation
T_values = simulate_sprayer()

# Define additional parameters
spray_angles = np.array([7 * np.pi / 18])
flow_rates = np.array([1083])
fixed_x_values = np.array([-150, 0, 150])

# Plot results and print the thickness table
plot_results(np.arange(0, 100.01, 0.01), T_values, fixed_x_values)
plot_thickness_variation(T_values, fixed_x_values, spray_angles, flow_rates)
print_thickness_table(T_values, fixed_x_values, spray_angles, flow_rates)
```



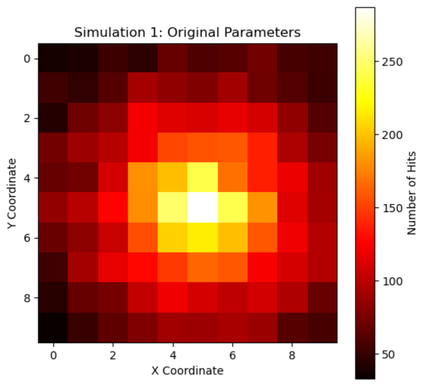


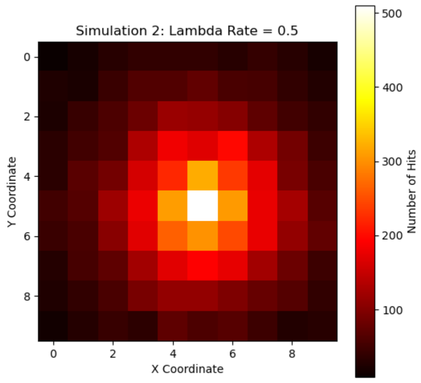

```
Thickness at Origin (x, y = (0, 0)) at the end of the simulation:

   Spray Angle (deg)70.0
--------------------------------------------------
-150                                0.00
0                                   0.00
150                                 0.01
```

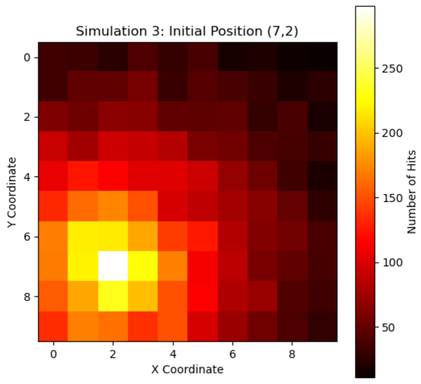
In [ ]:

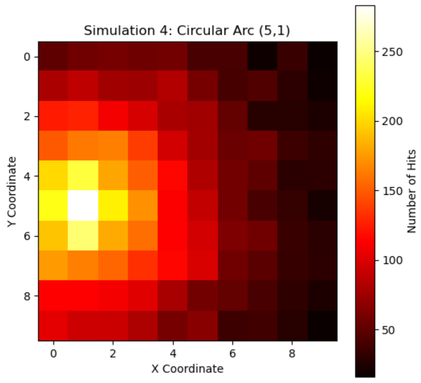
In [2]:

```python
import numpy as np
import matplotlib.pyplot as plt

# Define the simulate_sprayer function
def simulate_sprayer(lambda_rate, grid_size, num_passes, initial_x, initial_y):
    G = np.zeros((grid_size, grid_size))

    for k in range(num_passes):
        for i in range(grid_size):
            for j in range(grid_size):
                d = np.sqrt((initial_x - i) ** 2 + (initial_y - j) ** 2)
                p = lambda_rate * np.exp(-lambda_rate * d)
                beta = np.random.rand()

                if beta < p:
                    G[i, j] += 1

    return G

# Define a function to add a circular arc to the grid matrix
def add_circular_arc(grid_matrix, initial_x, initial_y, radius, num_points):
    theta = np.linspace(0, np.pi / 2, num_points)
    x_arc = np.round(radius * np.cos(theta) + initial_x).astype(int)
    y_arc = np.round(radius * np.sin(theta) + initial_y).astype(int)

    for x, y in zip(x_arc, y_arc):
        grid_matrix[x, y] += 1

    return grid_matrix

# Define a function to plot the grid matrix
def plot_grid_matrix(grid_matrix, title):
    plt.figure(figsize=(6, 6))
    plt.imshow(grid_matrix, cmap='hot', interpolation='nearest')
    plt.colorbar(label='Number of Hits')
    plt.title(title)
    plt.xlabel('X Coordinate')
    plt.ylabel('Y Coordinate')
    plt.show()

# Define simulation parameters
lambda_rate = 0.3
grid_size = 10
num_passes = 1000
initial_x, initial_y = 5, 5

# Perform and plot three different simulations
# Simulation 1: Original parameters
grid_matrix1 = simulate_sprayer(lambda_rate, grid_size, num_passes, initial_x, initial_y)
plot_grid_matrix(grid_matrix1, 'Simulation 1: Original Parameters')

# Simulation 2: Different lambda_rate
lambda_rate_2 = 0.5
grid_matrix2 = simulate_sprayer(lambda_rate_2, grid_size, num_passes, initial_x, initial_y)
plot_grid_matrix(grid_matrix2, 'Simulation 2: Lambda Rate = 0.5')

# Simulation 3: Different initial position
initial_x_3, initial_y_3 = 7, 2
grid_matrix3 = simulate_sprayer(lambda_rate, grid_size, num_passes, initial_x_3, initial_y_3)
plot_grid_matrix(grid_matrix3, 'Simulation 3: Initial Position (7,2)')

# Simulation 4: Circular arc
initial_x_4, initial_y_4 = 5, 1
radius_4 = 4
num_points_4 = 100
grid_matrix4 = simulate_sprayer(lambda_rate, grid_size, num_passes, initial_x_4, initial_y_4)
grid_matrix4 = add_circular_arc(grid_matrix4, initial_x_4, initial_y_4, radius_4, num_points_4)
plot_grid_matrix(grid_matrix4, 'Simulation 4: Circular Arc (5,1)')

# Show all plots
plt.show()
```









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E. Data Analytics

Code/Formulas used for data analytics

The team used Excel Formulas and Python to analyze and conduct data analysis for this activity. Appendix A represents a better understanding of the formulas used in excel to conduct data analysis. The following table represents a snipped of all the excel formulas used in order to conduct this activity, along with Python Code to perform analytics. Similar to the previous section, python was coded and run in Jupyter Notebook.

	Mean	Probablity of Failure	Residual	Residual Analysis
Assembly loose or misaligned	=AVERAGE(E2:E9)	=COUNT(E2:E9)/COUNT($E$2:$E$236)	=$B$240-B239	=D239*C239
Battery disfunction	=AVERAGE(E10:E70)	=COUNT(E10:E70)/COUNT($E$2:$E$236)	=$B$240-B240	=D240*C240
Electrical disfunction	=AVERAGE(E71:E81)	=COUNT(E71:E81)/COUNT($E$2:$E$236)	=$B$240-B241	=D241*C241
Left handle cracked or split	=AVERAGE(E82:E106)	=COUNT(E82:E106)/COUNT($E$2:$E$236)	=$B$240-B242	=D242*C242
Left upper casing cracked or split	=AVERAGE(E107:E129)	=COUNT(E107:E129)/COUNT($E$2:$E$236)	=$B$240-B243	=D243*C243
Nozzle cracked or split	=AVERAGE(E130:E131)	=COUNT(E130:E131)/COUNT($E$2:$E$236)	=$B$240-B244	=D244*C244
Right handle cracked or split	=AVERAGE(E132:E156)	=COUNT(E132:E156)/COUNT($E$2:$E$236)	=$B$240-B245	=D245*C245
Right upper casing cracked or split	=AVERAGE(E157:E190)	=COUNT(E157:E190)/COUNT($E$2:$E$236)	=$B$240-B246	=D246*C246
Trigger cracked or split	=AVERAGE(E191:E219)	=COUNT(E191:E219)/COUNT($E$2:$E$236)	=$B$240-B247	=D247*C247
Trigger loose or misaligned	=AVERAGE(E220:E236)	=COUNT(E220:E236)/COUNT($E$2:$E$236)	=$B$240-B248	=D248*C248

Mean (Days between Failures)	=AVERAGE(E2:E236)
Std (Days between Failures)	=STDEV(E2:E236)
NormCDF (xbar, xbar, std)	=NORM.DIST(B250,B250,B251,TRUE)

PDF	CDF
=NORM.DIST(E2:E236,B250,B251,FALSE)	=NORM.DIST(E2:E236,B250,B251,TRUE)

Sprayer fails before 600 days	=NORM.DIST(600,B253,B254,TRUE)
Sprayer lasts more than 600 days	=1-NORM.DIST(600,B253,B254,TRUE)

Python code for Predicted vs Reported values:

```
# importing libraries
import numpy as np
import csv
from scipy.stats import norm
import matplotlib.pyplot as plt
```

```
# name of input file
INPUT_FILE_NAME = 'Book1.csv'
# initializing mean and standard deviation
MEAN = 483.2553191
STD = 195.5553
# initializing list to store csv values
durations = []
# reading csv
with open(INPUT_FILE_NAME, mode = 'r') as file:
input_csv = csv.reader(file)
# skipping the header
next(input_csv)
for lines in input_csv:
durations.append(int(lines[0]))
# converting durations list to numpy array
durations = np.array(durations)
y_fit = norm.cdf(durations, loc=MEAN, scale=STD)
y_actual = np.zeros_like(durations, dtype=float)
for i in range(len(durations)):
y_actual[i] = np.sum(durations[durations < durations[i]]) / np.sum
# Plot the results
plt.plot(durations, y_fit,'.', label='predicted')
plt.plot(durations, y_actual, '.', label='reported')
plt.title('Predicted vs Reported values')
plt.xlabel('Lifespan (Days)')
plt.ylabel('Proportion of failures')
plt.legend()
plt.show()
```

Analysis and results

According to the data analysis the predicted lifespan of the sprayer is roughly 484 days or 1.3 years. This suggests that there is a 72.5% probability of the sprayer lasting atleast for 1.3 years on average. The most common cause of failure is the cracking or splitting of the right upper casing, which accounts for 14.4% of the total failures. This could be due to the stress or pressure applied to the casing during use or storage. Another major cause of failure is the splitting of the plastic trigger or shells, which accounts for 57.9% of the cumulative failures. This could be due to the material or thickness of the plastic parts, which may not be suitable for the sprayer's function or environment. A cracked trigger or a cracked outer shell would affect the overall performance of the sprayer and would hinder with the usability of the sprayer, as well as compromise the safety of the user.

To improve the reliability and durability of the sprayer, the most beneficial dimension to change would be the material or thickness of the outer shells and the trigger. By opting for a durable and robust material like metal or rubber the sprayer would be able to handle increased stress and pressure thereby minimizing the likelihood of cracks or splits. This modification would enhance customer satisfaction and experience for the product, while reducing the chances of potential leaks, breaks or cause damage to the user. Alternatively, the design or shape of the casing, trigger, or shells could be modified to reduce the stress points or increase the stability of the sprayer.

The assumptions were that the dates were recorded correctly, and that the failures were accurate and correctly attributed to the corresponding products. On the contrary, the sources of error which we think would cause errors include data entry errors, incomplete information, or missing context. For example, some data entries could be missing, incorrect, or duplicated, which could skew the statistics or calculations. Some information could be incomplete, such

as the reason or mode of failure, the frequency or intensity of use, or the environmental conditions of the sprayer. Some contexts could be missing, such as the user feedback, the warranty or return policy, or the quality control or testing procedures of the sprayer. These factors could influence the lifespan, likelihood, or cause of failure of the sprayer, when conducting data analysis for insights.

To improve the quality or precision of the analysis, additional information that would be helpful is data quality and quantity. Data quality refers to how accurate, comprehensive, consistent and relevant the information is. On the other hand, data quantity relates to the volume, variety and frequency of available data. By ensuring that the data that is provided or collected is reliable, and comprehensive, the process of analysis can be done more precisely and thus yield more robust results, thus providing better insights into the sprayer. A total of 9 rows were removed from the data set, the following table provides the list of data removed from the analysis.

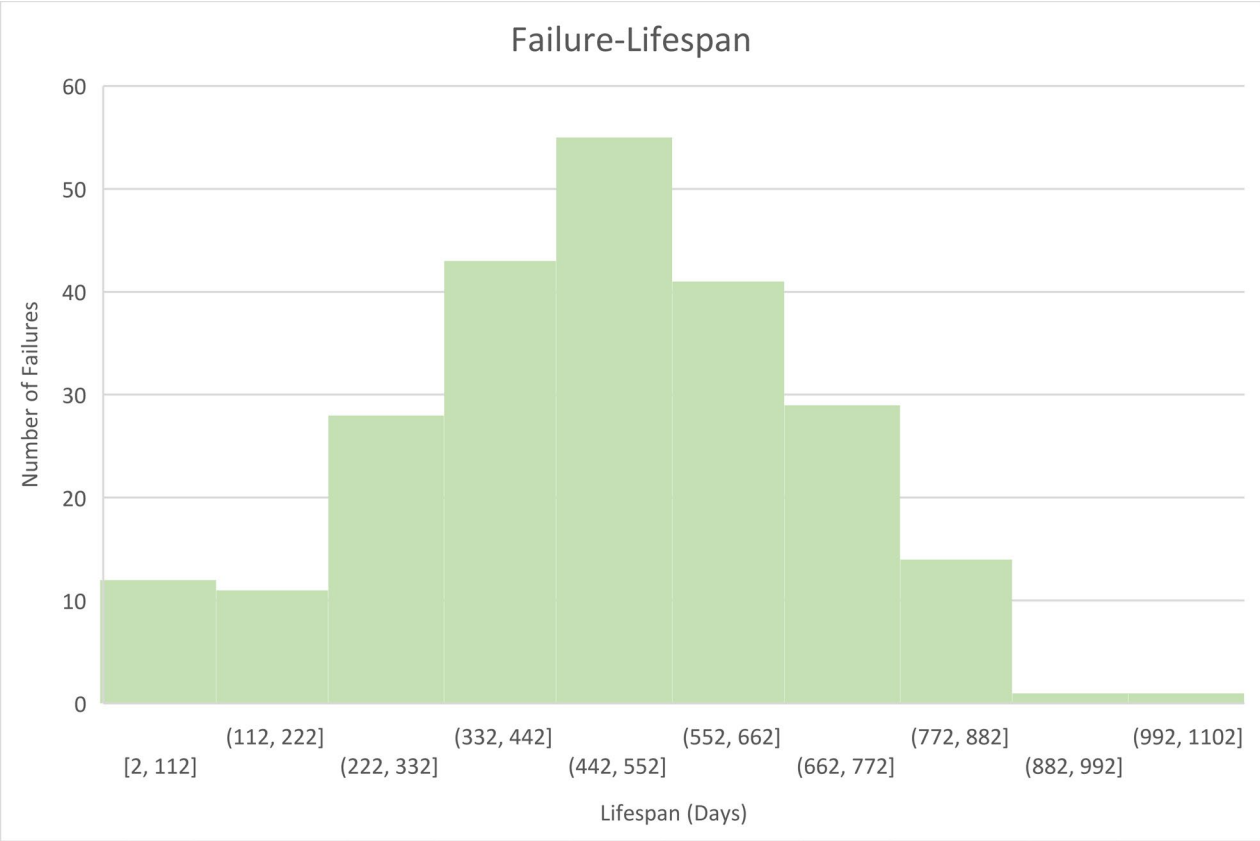
ID	Product	Reported Date	Sell Date	Failure Type	Addressed	Usage
9	HH-SPRAYER-ONEP	20-Oct-19	20-Oct-19	Trigger loose or misaligned	Provided new product	Did not report
183	HH-SPRAYER-ONEP	10-Sep-19	10-Sep-19	Trigger loose or misaligned	Provided new product	Did not report
187	HH-SPRAYER-ONEP	14-Aug-18	14-Aug-18	Assembly loose or misaligned	Provided new product	Agriculture
194	HH-SPRAYER-ONEP	22-Sep-20	22-Sep-20	Trigger loose or misaligned	Refunded purchase	Contractor
225	HH-SPRAYER-ONEP	7-Jun-17	7-Jun-17	Assembly loose or misaligned	Provided new product	Contractor
7	HH-SPRAYER-ONEP	3-Feb-20	17-Oct-18	missing	Provided new product	Did not report
131	HH-SPRAYER-ONEP	22-Aug-20	24-Dec-18	missing	Provided new product	Maintenance and Sanitation
196	HH-SPRAYER-ONEP	3-Jan-22	20-Nov-20	missing	Provided new product	Contractor
228	HH-SPRAYER-ONEP	21-Jul-22	6-Oct-20	missing	Refunded purchase	missing

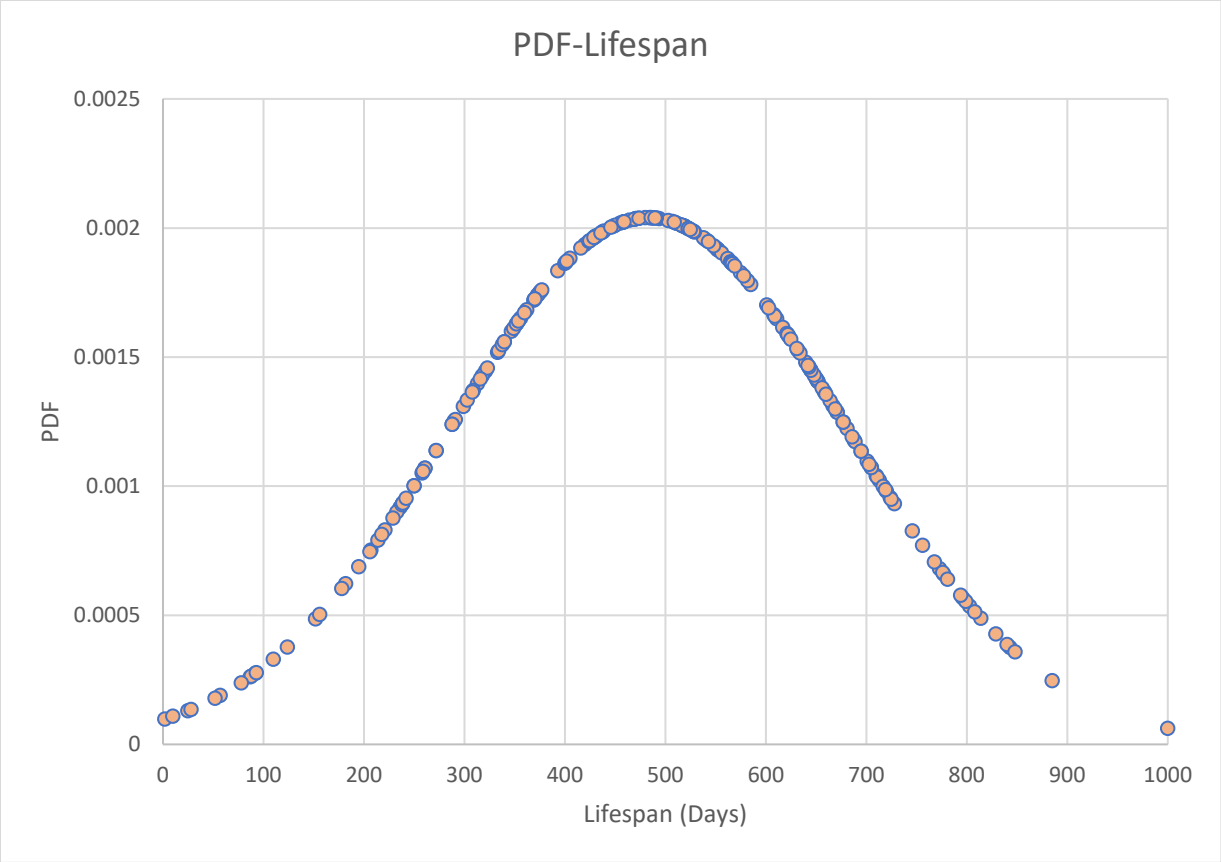
	Mean	Probability of Failure	Residual	Residual Analysis
Assembly loose or misaligned	174.5	0.034042553	471.9918	16.06780607
Battery disfunction	646.4918	0.259574468	0	0
Electrical disfunction	637.2727	0.046808511	9.219076	0.431531217
Left handle cracked or split	454.08	0.106382979	192.4118	20.46934077
Left upper casing cracked or split	450.4783	0.09787234	196.0135	19.18430415
Nozzle cracked or split	410	0.008510638	236.4918	2.012696198
Right handle cracked or split	377.08	0.106382979	269.4118	28.66083014
Right upper casing cracked or split	381.8529	0.144680851	264.6389	38.28817579
Trigger cracked or split	632.7586	0.123404255	13.73318	1.694733171
Trigger loose or misaligned	142.9412	0.072340426	503.5506	36.42706662

Mean (Days between Failures)	483.2553
Std (Days between Failures)	195.5553
NormCDF (xbar, xbar, std)	0.5

Sprayer fails before 600 days	0.724743
Sprayer lasts more than 600 days	0.275257

Plots for data analytics




PDF-Lifespan
0.0025
0.002
0.0015
0.001
0.0005
0
PDF
0
100
200
300
400
500
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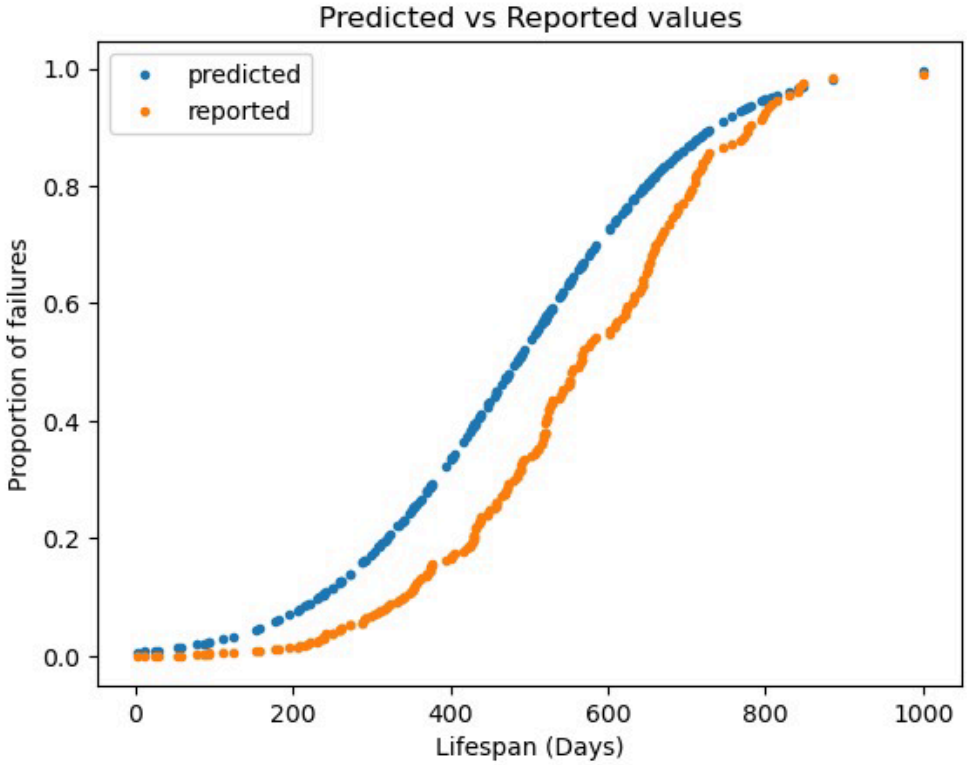
900
1000
Lifespan (Days)


Predicted vs Reported values
predicted
reported
1.0
0.8
0.6
0.4
0.2
0.0
Proportion of failures
0
200
400
600
800
1000
Lifespan (Days)

APPENDIX A

ID	Product	Reported Date	Sell Date	Days betweeen Failure	Failure Type
4	HH-SPRAYER-ONEP	44274	44016	=C2-D2	Assembly loose or misaligned
7	HH-SPRAYER-ONEP	43070	43045	=C3-D3	Assembly loose or misaligned
34	HH-SPRAYER-ONEP	44202	44078	=C4-D4	Assembly loose or misaligned
79	HH-SPRAYER-ONEP	43544	43392	=C5-D5	Assembly loose or misaligned
84	HH-SPRAYER-ONEP	43906	43904	=C6-D6	Assembly loose or misaligned
183	HH-SPRAYER-ONEP	43849	43610	=C7-D7	Assembly loose or misaligned
218	HH-SPRAYER-ONEP	43387	43012	=C8-D8	Assembly loose or misaligned
224	HH-SPRAYER-ONEP	43130	42909	=C9-D9	Assembly loose or misaligned
6	HH-SPRAYER-ONEP	43539	42762	=C10-D10	Battery disfunction
14	HH-SPRAYER-ONEP	44412	43745	=C11-D11	Battery disfunction
18	HH-SPRAYER-ONEP	44116	43302	=C12-D12	Battery disfunction
23	HH-SPRAYER-ONEP	44234	43583	=C13-D13	Battery disfunction
25	HH-SPRAYER-ONEP	43774	43018	=C14-D14	Battery disfunction
27	HH-SPRAYER-ONEP	44812	44124	=C15-D15	Battery disfunction
31	HH-SPRAYER-ONEP	44813	43970	=C16-D16	Battery disfunction
33	HH-SPRAYER-ONEP	44191	43580	=C17-D17	Battery disfunction
36	HH-SPRAYER-ONEP	43859	43207	=C18-D18	Battery disfunction
39	HH-SPRAYER-ONEP	44252	43506	=C19-D19	Battery disfunction
46	HH-SPRAYER-ONEP	43420	42768	=C20-D20	Battery disfunction
47	HH-SPRAYER-ONEP	44632	43943	=C21-D21	Battery disfunction
48	HH-SPRAYER-ONEP	43582	42786	=C22-D22	Battery disfunction
50	HH-SPRAYER-ONEP	44527	43527	=C23-D23	Battery disfunction
52	HH-SPRAYER-ONEP	44407	43748	=C24-D24	Battery disfunction
58	HH-SPRAYER-ONEP	44110	43651	=C25-D25	Battery disfunction
62	HH-SPRAYER-ONEP	44626	44073	=C26-D26	Battery disfunction
63	HH-SPRAYER-ONEP	44475	43774	=C27-D27	Battery disfunction
66	HH-SPRAYER-ONEP	43452	42781	=C28-D28	Battery disfunction
67	HH-SPRAYER-ONEP	44196	43551	=C29-D29	Battery disfunction
69	HH-SPRAYER-ONEP	44824	44104	=C30-D30	Battery disfunction
73	HH-SPRAYER-ONEP	44026	43606	=C31-D31	Battery disfunction
74	HH-SPRAYER-ONEP	43810	42981	=C32-D32	Battery disfunction
78	HH-SPRAYER-ONEP	43397	42926	=C33-D33	Battery disfunction
80	HH-SPRAYER-ONEP	44394	43684	=C34-D34	Battery disfunction
82	HH-SPRAYER-ONEP	43759	43252	=C35-D35	Battery disfunction
94	HH-SPRAYER-ONEP	44202	43552	=C36-D36	Battery disfunction
97	HH-SPRAYER-ONEP	44564	44045	=C37-D37	Battery disfunction
102	HH-SPRAYER-ONEP	43530	42835	=C38-D38	Battery disfunction
104	HH-SPRAYER-ONEP	43401	42849	=C39-D39	Battery disfunction
107	HH-SPRAYER-ONEP	43636	43165	=C40-D40	Battery disfunction
108	HH-SPRAYER-ONEP	44316	43476	=C41-D41	Battery disfunction
111	HH-SPRAYER-ONEP	43920	43368	=C42-D42	Battery disfunction
113	HH-SPRAYER-ONEP	44115	43387	=C43-D43	Battery disfunction
126	HH-SPRAYER-ONEP	44071	43268	=C44-D44	Battery disfunction
127	HH-SPRAYER-ONEP	43351	42799	=C45-D45	Battery disfunction
132	HH-SPRAYER-ONEP	43654	43160	=C46-D46	Battery disfunction
136	HH-SPRAYER-ONEP	43646	43106	=C47-D47	Battery disfunction
140	HH-SPRAYER-ONEP	44482	43834	=C48-D48	Battery disfunction
144	HH-SPRAYER-ONEP	43329	42810	=C49-D49	Battery disfunction
146	HH-SPRAYER-ONEP	44820	44115	=C50-D50	Battery disfunction
150	HH-SPRAYER-ONEP	43806	43033	=C51-D51	Battery disfunction
155	HH-SPRAYER-ONEP	44387	43619	=C52-D52	Battery disfunction
167	HH-SPRAYER-ONEP	43622	42846	=C53-D53	Battery disfunction
171	HH-SPRAYER-ONEP	43539	42954	=C54-D54	Battery disfunction
181	HH-SPRAYER-ONEP	44100	43442	=C55-D55	Battery disfunction
189	HH-SPRAYER-ONEP	44853	44045	=C56-D56	Battery disfunction
192	HH-SPRAYER-ONEP	44876	44163	=C57-D57	Battery disfunction
195	HH-SPRAYER-ONEP	44596	44137	=C58-D58	Battery disfunction
205	HH-SPRAYER-ONEP	44645	44103	=C59-D59	Battery disfunction
208	HH-SPRAYER-ONEP	43637	43110	=C60-D60	Battery disfunction
209	HH-SPRAYER-ONEP	43425	43073	=C61-D61	Battery disfunction
210	HH-SPRAYER-ONEP	44544	43977	=C62-D62	Battery disfunction
214	HH-SPRAYER-ONEP	44761	43913	=C63-D63	Battery disfunction
216	HH-SPRAYER-ONEP	43658	43147	=C64-D64	Battery disfunction
217	HH-SPRAYER-ONEP	43356	42853	=C65-D65	Battery disfunction
220	HH-SPRAYER-ONEP	43326	42776	=C66-D66	Battery disfunction
222	HH-SPRAYER-ONEP	44332	43700	=C67-D67	Battery disfunction
228	HH-SPRAYER-ONEP	44752	44112	=C68-D68	Battery disfunction
232	HH-SPRAYER-ONEP	43598	42799	=C69-D69	Battery disfunction
235	HH-SPRAYER-ONEP	43939	43469	=C70-D70	Battery disfunction
10	HH-SPRAYER-ONEP	44690	44026	=C71-D71	Electrical disfunction
15	HH-SPRAYER-ONEP	44480	43837	=C72-D72	Electrical disfunction
26	HH-SPRAYER-ONEP	44520	43864	=C73-D73	Electrical disfunction
55	HH-SPRAYER-ONEP	44163	43482	=C74-D74	Electrical disfunction
87	HH-SPRAYER-ONEP	43656	43011	=C75-D75	Electrical disfunction
148	HH-SPRAYER-ONEP	43714	43106	=C76-D76	Electrical disfunction
193	HH-SPRAYER-ONEP	44112	43401	=C77-D77	Electrical disfunction
219	HH-SPRAYER-ONEP	43765	43156	=C78-D78	Electrical disfunction
227	HH-SPRAYER-ONEP	43835	43218	=C79-D79	Electrical disfunction
230	HH-SPRAYER-ONEP	44656	44055	=C80-D80	Electrical disfunction
234	HH-SPRAYER-ONEP	44137	43562	=C81-D81	Electrical disfunction
13	HH-SPRAYER-ONEP	44044	43506	=C82-D82	Left handle cracked or split
17	HH-SPRAYER-ONEP	44113	43591	=C83-D83	Left handle cracked or split
37	HH-SPRAYER-ONEP	44099	43644	=C84-D84	Left handle cracked or split
38	HH-SPRAYER-ONEP	44572	43901	=C85-D85	Left handle cracked or split
42	HH-SPRAYER-ONEP	43495	43064	=C86-D86	Left handle cracked or split
49	HH-SPRAYER-ONEP	44384	43899	=C87-D87	Left handle cracked or split
65	HH-SPRAYER-ONEP	43869	43515	=C88-D88	Left handle cracked or split
70	HH-SPRAYER-ONEP	43396	42991	=C89-D89	Left handle cracked or split
71	HH-SPRAYER-ONEP	43921	43574	=C90-D90	Left handle cracked or split

81	HH-SPRAYER-ONEP	44342	43986	=C91-D91	Left handle cracked or split
91	HH-SPRAYER-ONEP	43464	43257	=C92-D92	Left handle cracked or split
100	HH-SPRAYER-ONEP	44162	43651	=C93-D93	Left handle cracked or split
103	HH-SPRAYER-ONEP	44243	43759	=C94-D94	Left handle cracked or split
110	HH-SPRAYER-ONEP	44603	44171	=C95-D95	Left handle cracked or split
112	HH-SPRAYER-ONEP	44021	43530	=C96-D96	Left handle cracked or split
147	HH-SPRAYER-ONEP	44540	44110	=C97-D97	Left handle cracked or split
153	HH-SPRAYER-ONEP	44035	43606	=C98-D98	Left handle cracked or split
157	HH-SPRAYER-ONEP	43333	42767	=C99-D99	Left handle cracked or split
165	HH-SPRAYER-ONEP	44406	43968	=C100-D100	Left handle cracked or split
178	HH-SPRAYER-ONEP	43468	42834	=C101-D101	Left handle cracked or split
196	HH-SPRAYER-ONEP	43812	43256	=C102-D102	Left handle cracked or split
203	HH-SPRAYER-ONEP	43545	43129	=C103-D103	Left handle cracked or split
211	HH-SPRAYER-ONEP	43232	42758	=C104-D104	Left handle cracked or split
212	HH-SPRAYER-ONEP	43435	43163	=C105-D105	Left handle cracked or split
223	HH-SPRAYER-ONEP	43985	43537	=C106-D106	Left handle cracked or split
5	HH-SPRAYER-ONEP	43972	43351	=C107-D107	Left upper casing cracked or split
16	HH-SPRAYER-ONEP	43657	43095	=C108-D108	Left upper casing cracked or split
20	HH-SPRAYER-ONEP	44577	44149	=C109-D109	Left upper casing cracked or split
24	HH-SPRAYER-ONEP	43706	43276	=C110-D110	Left upper casing cracked or split
43	HH-SPRAYER-ONEP	43454	43017	=C111-D111	Left upper casing cracked or split
59	HH-SPRAYER-ONEP	43358	42776	=C112-D112	Left upper casing cracked or split
89	HH-SPRAYER-ONEP	44277	43956	=C113-D113	Left upper casing cracked or split
95	HH-SPRAYER-ONEP	44648	44100	=C114-D114	Left upper casing cracked or split
114	HH-SPRAYER-ONEP	43860	43344	=C115-D115	Left upper casing cracked or split
123	HH-SPRAYER-ONEP	43829	43647	=C116-D116	Left upper casing cracked or split
125	HH-SPRAYER-ONEP	44070	43677	=C117-D117	Left upper casing cracked or split
133	HH-SPRAYER-ONEP	43248	42958	=C118-D118	Left upper casing cracked or split
138	HH-SPRAYER-ONEP	44636	44013	=C119-D119	Left upper casing cracked or split
141	HH-SPRAYER-ONEP	44199	43749	=C120-D120	Left upper casing cracked or split
163	HH-SPRAYER-ONEP	43599	43281	=C121-D121	Left upper casing cracked or split
164	HH-SPRAYER-ONEP	43479	43079	=C122-D122	Left upper casing cracked or split
174	HH-SPRAYER-ONEP	43381	43068	=C123-D123	Left upper casing cracked or split
185	HH-SPRAYER-ONEP	43728	43304	=C124-D124	Left upper casing cracked or split
188	HH-SPRAYER-ONEP	43565	43140	=C125-D125	Left upper casing cracked or split
191	HH-SPRAYER-ONEP	44516	44028	=C126-D126	Left upper casing cracked or split
197	HH-SPRAYER-ONEP	44512	43843	=C127-D127	Left upper casing cracked or split
198	HH-SPRAYER-ONEP	43630	43065	=C128-D128	Left upper casing cracked or split
200	HH-SPRAYER-ONEP	43353	42977	=C129-D129	Left upper casing cracked or split
159	HH-SPRAYER-ONEP	43370	42841	=C130-D130	Nozzle cracked or split
186	HH-SPRAYER-ONEP	44018	43727	=C131-D131	Nozzle cracked or split
1	HH-SPRAYER-ONEP	44087	43788	=C132-D132	Right handle cracked or split
3	HH-SPRAYER-ONEP	43730	43552	=C133-D133	Right handle cracked or split
11	HH-SPRAYER-ONEP	44584	44145	=C134-D134	Right handle cracked or split
44	HH-SPRAYER-ONEP	43945	43636	=C135-D135	Right handle cracked or split
56	HH-SPRAYER-ONEP	44408	44120	=C136-D136	Right handle cracked or split
90	HH-SPRAYER-ONEP	43884	43511	=C137-D137	Right handle cracked or split
96	HH-SPRAYER-ONEP	44379	43861	=C138-D138	Right handle cracked or split
99	HH-SPRAYER-ONEP	44294	43945	=C139-D139	Right handle cracked or split
105	HH-SPRAYER-ONEP	43589	43109	=C140-D140	Right handle cracked or split
116	HH-SPRAYER-ONEP	43602	43279	=C141-D141	Right handle cracked or split
118	HH-SPRAYER-ONEP	43712	43462	=C142-D142	Right handle cracked or split
119	HH-SPRAYER-ONEP	43894	43622	=C143-D143	Right handle cracked or split
124	HH-SPRAYER-ONEP	43527	43010	=C144-D144	Right handle cracked or split
130	HH-SPRAYER-ONEP	43203	42773	=C145-D145	Right handle cracked or split
143	HH-SPRAYER-ONEP	44326	43974	=C146-D146	Right handle cracked or split
156	HH-SPRAYER-ONEP	43350	42921	=C147-D147	Right handle cracked or split
158	HH-SPRAYER-ONEP	43897	43594	=C148-D148	Right handle cracked or split
162	HH-SPRAYER-ONEP	43782	43204	=C149-D149	Right handle cracked or split
172	HH-SPRAYER-ONEP	43639	43118	=C150-D150	Right handle cracked or split
173	HH-SPRAYER-ONEP	44447	44159	=C151-D151	Right handle cracked or split
175	HH-SPRAYER-ONEP	44317	43944	=C152-D152	Right handle cracked or split
179	HH-SPRAYER-ONEP	43555	43294	=C153-D153	Right handle cracked or split
190	HH-SPRAYER-ONEP	43625	43106	=C154-D154	Right handle cracked or split
204	HH-SPRAYER-ONEP	43556	43155	=C155-D155	Right handle cracked or split
233	HH-SPRAYER-ONEP	44472	44095	=C156-D156	Right handle cracked or split
9	HH-SPRAYER-ONEP	44284	43915	=C157-D157	Right upper casing cracked or split
19	HH-SPRAYER-ONEP	43037	42801	=C158-D158	Right upper casing cracked or split
22	HH-SPRAYER-ONEP	43117	42763	=C159-D159	Right upper casing cracked or split
29	HH-SPRAYER-ONEP	43335	42933	=C160-D160	Right upper casing cracked or split
30	HH-SPRAYER-ONEP	43482	43221	=C161-D161	Right upper casing cracked or split
41	HH-SPRAYER-ONEP	44441	43922	=C162-D162	Right upper casing cracked or split
51	HH-SPRAYER-ONEP	43505	43255	=C163-D163	Right upper casing cracked or split
53	HH-SPRAYER-ONEP	43467	43097	=C164-D164	Right upper casing cracked or split
54	HH-SPRAYER-ONEP	44242	43902	=C165-D165	Right upper casing cracked or split
61	HH-SPRAYER-ONEP	43872	43639	=C166-D166	Right upper casing cracked or split
64	HH-SPRAYER-ONEP	44161	43923	=C167-D167	Right upper casing cracked or split
76	HH-SPRAYER-ONEP	43215	42751	=C168-D168	Right upper casing cracked or split
83	HH-SPRAYER-ONEP	43381	43048	=C169-D169	Right upper casing cracked or split
86	HH-SPRAYER-ONEP	44469	44161	=C170-D170	Right upper casing cracked or split
101	HH-SPRAYER-ONEP	43934	43368	=C171-D171	Right upper casing cracked or split
109	HH-SPRAYER-ONEP	43420	43206	=C172-D172	Right upper casing cracked or split
115	HH-SPRAYER-ONEP	44271	43783	=C173-D173	Right upper casing cracked or split
137	HH-SPRAYER-ONEP	43331	42873	=C174-D174	Right upper casing cracked or split
142	HH-SPRAYER-ONEP	43352	43018	=C175-D175	Right upper casing cracked or split
168	HH-SPRAYER-ONEP	44180	43664	=C176-D176	Right upper casing cracked or split
169	HH-SPRAYER-ONEP	43733	43363	=C177-D177	Right upper casing cracked or split
170	HH-SPRAYER-ONEP	43182	42844	=C178-D178	Right upper casing cracked or split
177	HH-SPRAYER-ONEP	44481	43958	=C179-D179	Right upper casing cracked or split
180	HH-SPRAYER-ONEP	44478	43969	=C180-D180	Right upper casing cracked or split

184	HH-SPRAYER-ONEP	43436	43120	=C181-D181	Right upper casing cracked or split
194	HH-SPRAYER-ONEP	43974	43505	=C182-D182	Right upper casing cracked or split
199	HH-SPRAYER-ONEP	44457	44021	=C183-D183	Right upper casing cracked or split
202	HH-SPRAYER-ONEP	42915	42805	=C184-D184	Right upper casing cracked or split
206	HH-SPRAYER-ONEP	44594	44120	=C185-D185	Right upper casing cracked or split
207	HH-SPRAYER-ONEP	44145	43659	=C186-D186	Right upper casing cracked or split
213	HH-SPRAYER-ONEP	43275	42751	=C187-D187	Right upper casing cracked or split
225	HH-SPRAYER-ONEP	44474	43984	=C188-D188	Right upper casing cracked or split
229	HH-SPRAYER-ONEP	44394	43948	=C189-D189	Right upper casing cracked or split
231	HH-SPRAYER-ONEP	44112	43873	=C190-D190	Right upper casing cracked or split
2	HH-SPRAYER-ONEP	43991	43348	=C191-D191	Trigger cracked or split
8	HH-SPRAYER-ONEP	44365	43705	=C192-D192	Trigger cracked or split
12	HH-SPRAYER-ONEP	43875	43332	=C193-D193	Trigger cracked or split
21	HH-SPRAYER-ONEP	44877	44153	=C194-D194	Trigger cracked or split
32	HH-SPRAYER-ONEP	43973	43611	=C195-D195	Trigger cracked or split
40	HH-SPRAYER-ONEP	43944	43416	=C196-D196	Trigger cracked or split
60	HH-SPRAYER-ONEP	44222	43428	=C197-D197	Trigger cracked or split
68	HH-SPRAYER-ONEP	43833	43191	=C198-D198	Trigger cracked or split
72	HH-SPRAYER-ONEP	43708	42991	=C199-D199	Trigger cracked or split
75	HH-SPRAYER-ONEP	43278	42755	=C200-D200	Trigger cracked or split
77	HH-SPRAYER-ONEP	44405	43838	=C201-D201	Trigger cracked or split
85	HH-SPRAYER-ONEP	43866	43180	=C202-D202	Trigger cracked or split
92	HH-SPRAYER-ONEP	43423	42801	=C203-D203	Trigger cracked or split
93	HH-SPRAYER-ONEP	43777	43052	=C204-D204	Trigger cracked or split
98	HH-SPRAYER-ONEP	43679	43154	=C205-D205	Trigger cracked or split
106	HH-SPRAYER-ONEP	44590	43987	=C206-D206	Trigger cracked or split
122	HH-SPRAYER-ONEP	44545	44086	=C207-D207	Trigger cracked or split
128	HH-SPRAYER-ONEP	43718	42999	=C208-D208	Trigger cracked or split
129	HH-SPRAYER-ONEP	44172	43391	=C209-D209	Trigger cracked or split
131	HH-SPRAYER-ONEP	44231	43871	=C210-D210	Trigger cracked or split
134	HH-SPRAYER-ONEP	45072	44187	=C211-D211	Trigger cracked or split
135	HH-SPRAYER-ONEP	43941	43246	=C212-D212	Trigger cracked or split
139	HH-SPRAYER-ONEP	44728	44159	=C213-D213	Trigger cracked or split
145	HH-SPRAYER-ONEP	43596	42919	=C214-D214	Trigger cracked or split
154	HH-SPRAYER-ONEP	43694	43017	=C215-D215	Trigger cracked or split
160	HH-SPRAYER-ONEP	44012	43387	=C216-D216	Trigger cracked or split
166	HH-SPRAYER-ONEP	44673	44042	=C217-D217	Trigger cracked or split
176	HH-SPRAYER-ONEP	43812	43107	=C218-D218	Trigger cracked or split
215	HH-SPRAYER-ONEP	44178	43475	=C219-D219	Trigger cracked or split
28	HH-SPRAYER-ONEP	43878	43538	=C220-D220	Trigger loose or misaligned
35	HH-SPRAYER-ONEP	43395	43153	=C221-D221	Trigger loose or misaligned
45	HH-SPRAYER-ONEP	43686	43530	=C222-D222	Trigger loose or misaligned
57	HH-SPRAYER-ONEP	44115	43909	=C223-D223	Trigger loose or misaligned
88	HH-SPRAYER-ONEP	43803	43608	=C224-D224	Trigger loose or misaligned
117	HH-SPRAYER-ONEP	43041	42984	=C225-D225	Trigger loose or misaligned
120	HH-SPRAYER-ONEP	43136	42918	=C226-D226	Trigger loose or misaligned
121	HH-SPRAYER-ONEP	43654	43562	=C227-D227	Trigger loose or misaligned
149	HH-SPRAYER-ONEP	43540	43512	=C228-D228	Trigger loose or misaligned
151	HH-SPRAYER-ONEP	44072	43985	=C229-D229	Trigger loose or misaligned
152	HH-SPRAYER-ONEP	43095	43085	=C230-D230	Trigger loose or misaligned
161	HH-SPRAYER-ONEP	43215	42956	=C231-D231	Trigger loose or misaligned
182	HH-SPRAYER-ONEP	44224	44136	=C232-D232	Trigger loose or misaligned
187	HH-SPRAYER-ONEP	44260	44167	=C233-D233	Trigger loose or misaligned
201	HH-SPRAYER-ONEP	43015	42786	=C234-D234	Trigger loose or misaligned
221	HH-SPRAYER-ONEP	42842	42790	=C235-D235	Trigger loose or misaligned
226	HH-SPRAYER-ONEP	43152	43074	=C236-D236	Trigger loose or misaligned

	Mean	Probablity of Failure	Residual	Residual Analysis
Assembly loose or misaligned	=AVERAGE(E2:E9)	=COUNT(E2:E9)/COUNT($E$2:$E$236)	=$B$240-B239	=D239*C239
Battery disfunction	=AVERAGE(E10:E70)	=COUNT(E10:E70)/COUNT($E$2:$E$236)	=$B$240-B240	=D240*C240
Electrical disfunction	=AVERAGE(E71:E81)	=COUNT(E71:E81)/COUNT($E$2:$E$236)	=$B$240-B241	=D241*C241
Left handle cracked or split	=AVERAGE(E82:E106)	=COUNT(E82:E106)/COUNT($E$2:$E$236)	=$B$240-B242	=D242*C242
Left upper casing cracked or split	=AVERAGE(E107:E129)	=COUNT(E107:E129)/COUNT($E$2:$E$236)	=$B$240-B243	=D243*C243
Nozzle cracked or split	=AVERAGE(E130:E131)	=COUNT(E130:E131)/COUNT($E$2:$E$236)	=$B$240-B244	=D244*C244
Right handle cracked or split	=AVERAGE(E132:E156)	=COUNT(E132:E156)/COUNT($E$2:$E$236)	=$B$240-B245	=D245*C245
Right upper casing cracked or split	=AVERAGE(E157:E190)	=COUNT(E157:E190)/COUNT($E$2:$E$236)	=$B$240-B246	=D246*C246
Trigger cracked or split	=AVERAGE(E191:E219)	=COUNT(E191:E219)/COUNT($E$2:$E$236)	=$B$240-B247	=D247*C247
Trigger loose or misaligned	=AVERAGE(E220:E236)	=COUNT(E220:E236)/COUNT($E$2:$E$236)	=$B$240-B248	=D248*C248

Mean (Days between Failures)	=AVERAGE(E2:E236)
Std (Days between Failures)	=STDEV(E2:E236)
NormCDF (xbar, xbar, std)	=NORM.DIST(B250,B250,B251,TRUE)

PDF	CDF
=NORM.DIST(E2:E236,B250,B251,FALSE)	=NORM.DIST(E2:E236,B250,B251,TRUE)

Sprayer fails before 600 days	=NORM.DIST(600,B253,B254,TRUE)
Sprayer lasts more than 600 days	=1-NORM.DIST(600,B253,B254,TRUE)